

Comprehensive Notes: Systems & Iterative Methods

Echelon Form • Rank of Matrix • Consistency • Gauss Elimination • Gauss-Seidel Method

1. Row Echelon Form (REF) & RREF

Definition of Row Echelon Form (REF)

A matrix is in **Row Echelon Form (REF)** if it satisfies three structural conditions:

- **Zero Rows at Bottom:** All rows consisting entirely of zeros (if any) are located below all non-zero rows.
- **Leading Entry Position:** The first non-zero entry (called the *leading entry* or *pivot*) in any non-zero row is strictly to the right of the leading entry of the row above it.
- **Zeros Below Pivots:** All entries in a column directly below a leading pivot are zero.

Reduced Row Echelon Form (RREF): Further requires that every leading pivot equals 1, and each leading pivot is the *only* non-zero entry in its column.

Standard Elementary Row Operations (EROs)

We perform EROs on a matrix to reduce it to Echelon form without altering its solution set:

1. **Row Interchange:** $R_i \leftrightarrow R_j$ — Swap row i and row j .
2. **Non-Zero Scalar Multiplication:** $R_i \rightarrow k \cdot R_i$ (where $k \neq 0$).
3. **Row Addition / Subtraction:** $R_i \rightarrow R_i + k \cdot R_j$ — Add a multiple of row j to row i .

2. Rank of a Matrix

The **Rank of a Matrix A** (denoted as $\rho(A)$ or $\text{rank}(A)$) is defined as:

- The **number of non-zero rows** in its Row Echelon Form.
- Equivalently, the maximum number of *linearly independent* row vectors or column vectors in A.

Fundamental Properties of Rank:

- For an $m \times n$ matrix A: $0 \leq \text{rank}(A) \leq \min(m, n)$.
- $\text{rank}(A) = 0$ if and only if A is a Zero Matrix.
- $\text{rank}(A) = \text{rank}(A^T)$.
- Elementary row operations preserve the rank of a matrix.

Example: Finding Matrix Rank via REF Reduction

Find the rank of matrix A :

$$A = \begin{bmatrix} 1 & -1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 4 & 0 & 4 & 8 \end{bmatrix}$$

Step 1: Apply $R_2 \rightarrow R_2 - 2R_1$ and $R_3 \rightarrow R_3 - 4R_1$

$$\begin{bmatrix} 1 & -1 & 2 & 3 \\ 0 & 4 & -4 & -4 \\ 0 & 4 & -4 & -4 \end{bmatrix}$$

Step 2: Apply $R_3 \rightarrow R_3 - R_2$ and scale $R_2 \rightarrow \frac{1}{4} R_2$

$$\begin{bmatrix} 1 & -1 & 2 & 3 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Result: There are **2 non-zero rows** in Echelon Form. Hence, $\text{rank}(A) = 2$.

3. Systems of Linear Equations in Three Variables

A system of three linear equations in three unknown variables (x, y, z) takes the standard form:

$$a_1x + b_1y + c_1z = d_1$$

$$a_2x + b_2y + c_2z = d_2$$

$$a_3x + b_3y + c_3z = d_3$$

In matrix notation: $AX = B$, where:

$$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad B = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

The **Augmented Matrix** combining coefficient matrix A and constant vector B is written as $[A \mid B]$.

4. Conditions for Consistency & Homogeneity

Let $n = 3$ be the number of unknown variables (x, y, z) .

A. Non-Homogeneous System ($AX = B$, where $B \neq 0$)

Rouché–Capelli Theorem Rules:

Condition	Rank Relationship	System Classification	Nature of Solution
$rank(A) = rank([A B]) = n$	$r = r' = 3$	Consistent & Independent	Exactly ONE Unique Solution
$rank(A) = rank([A B]) < n$	$r = r' < 3$	Consistent & Dependent	INFINITELY MANY Solutions
$rank(A) \neq rank([A B])$	$r \neq r'$	Inconsistent	NO Solution (Contradictory system)

B. Homogeneous System ($AX = 0$)

Since the zero vector is always in the column space, $rank(A) = rank([A|0])$ always holds. Thus, a homogeneous system is **always consistent**.

Determinant	Rank Condition	Solution Type
$ A \neq 0$ (Non-singular)	$rank(A) = 3$	Trivial Solution Only: $(x, y, z) = (0, 0, 0)$
$ A = 0$ (Singular)	$rank(A) < 3$	Non-Trivial Solutions: Infinitely many non-zero solutions

Example 1: Non-Homogeneous — Consistent with UNIQUE Solution

Solve the following system using Gauss Elimination:

$$x + y + z = 6$$

$$2x - y + z = 3$$

$$x + 2y + 3z = 14$$

Step 1: Write Augmented Matrix $[A|B]$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & -1 & 1 & 3 \\ 1 & 2 & 3 & 14 \end{array} \right]$$

Step 2: Reduce to Echelon Form

Apply $R_2 \rightarrow R_2 - 2R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -3 & -1 & -9 \\ 0 & 1 & 2 & 8 \end{array} \right]$$

Swap $R_2 \leftrightarrow R_3$ for a cleaner pivot:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & -3 & -1 & -9 \end{array} \right]$$

Apply $R_3 \rightarrow R_3 + 3R_2$:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & 5 & 15 \end{array} \right]$$

Step 3: Check Rank & Consistency

$\text{rank}(A) = 3$, $\text{rank}([A|B]) = 3$. Since $\text{rank}(A) = \text{rank}([A|B]) = n = 3$, the system has a **Unique Solution**.

Step 4: Back Substitution

- From Row 3: $5z = 15 \Rightarrow z = 3$
- From Row 2: $y + 2(3) = 8 \Rightarrow y + 6 = 8 \Rightarrow y = 2$
- From Row 1: $x + 2 + 3 = 6 \Rightarrow x + 5 = 6 \Rightarrow x = 1$

Final Solution: $(x, y, z) = (1, 2, 3)$

Example 2: Non-Homogeneous — Consistent with INFINITELY MANY Solutions

Solve the system:

$$\begin{aligned}x + 2y - z &= 3 \\2x + 4y + z &= 9 \\3x + 6y &= 12\end{aligned}$$

Step 1: Form Augmented Matrix $[A|B]$

$$\left| \begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 2 & 4 & 1 & 9 \\ 3 & 6 & 0 & 12 \end{array} \right|$$

Step 2: Reduce to Echelon Form

Apply $R_2 \rightarrow R_2 - 2R_1$ and $R_3 \rightarrow R_3 - 3R_1$:

$$\left| \begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 3 & 3 \end{array} \right|$$

Apply $R_3 \rightarrow R_3 - R_2$:

$$\left| \begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 0 \end{array} \right|$$

Step 3: Analyze Rank

$\text{rank}(A) = 2$, $\text{rank}([A|B]) = 2$. Since $\text{rank}(A) = \text{rank}([A|B]) = 2 < 3$, the system is **consistent with infinitely many solutions**.

Step 4: Back Substitution & Free Parameter

- From Row 2: $3z = 3 \Rightarrow z = 1$
- From Row 1: $x + 2y - 1 = 3 \Rightarrow x + 2y = 4$
- Let $y = k$ (where $k \in \mathbb{R}$ is a free variable): $x = 4 - 2k$

Parametric Solution: $(x, y, z) = (4 - 2k, k, 1)$ for any $k \in \mathbb{R}$.

Example 3: Non-Homogeneous — INCONSISTENT (No Solution)

Solve the system:

$$x + 2y + z = 4$$

$$2x + 4y + 2z = 8$$

$$3x + 6y + 3z = 10$$

Step 1: Form Augmented Matrix $[A|B]$

$$\left| \begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 2 & 4 & 2 & 8 \\ 3 & 6 & 3 & 10 \end{array} \right|$$

Step 2: Reduce to Echelon Form

Apply $R_2 \rightarrow R_2 - 2R_1$ and $R_3 \rightarrow R_3 - 3R_1$:

$$\left| \begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -2 \end{array} \right|$$

Swap $R_2 \leftrightarrow R_3$:

$$\left| \begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & 0 & 0 & -2 \\ 0 & 0 & 0 & 0 \end{array} \right|$$

Step 3: Analyze Rank & Conclusion

- $\text{rank}(A) = 1$ (coefficient part has 1 non-zero row).
- $\text{rank}([A|B]) = 2$ (augmented matrix has 2 non-zero rows).

Row 2 yields the equation $0x + 0y + 0z = -2$, which means $0 = -2$ (a physical contradiction).

Conclusion: Since $\text{rank}(A) \neq \text{rank}([A|B])$, the system is **INCONSISTENT** and has **NO SOLUTION**.

6. Gauss-Seidel Method (Iterative Method)

Theory & Working Principle

The **Gauss-Seidel Method** is an iterative technique used to solve a system of linear equations $AX = B$.

1. Strict Diagonal Dominance (SDD) Condition:

The system **must** be arranged such that the coefficient matrix **A** is *diagonally dominant*:

$$|a_{11}| > |b_1| + |c_1|$$

$$|b_2| > |a_2| + |c_2|$$

$$|c_3| > |a_3| + |b_3|$$

2. Iteration Formulas:

Express each variable in terms of the others:

$$x^{(k+1)} = 1/a_1 [d_1 - b_1 y^{(k)} - c_1 z^{(k)}]$$

$$y^{(k+1)} = 1/b_2 [d_2 - a_2 x^{(k+1)} - c_2 z^{(k)}] \text{ (uses new } x)$$

$$z^{(k+1)} = 1/c_3 [d_3 - a_3 x^{(k+1)} - b_3 y^{(k+1)}] \text{ (uses new } x \text{ and } y)$$

Key Feature: Unlike Jacobi's method, Gauss-Seidel **immediately uses newly computed values** within the same iteration step, leading to faster convergence.

Gauss-Seidel Example 1: Standard Diagonally Dominant System

Solve the following system starting with initial approximation $(x_0, y_0, z_0) = (0, 0, 0)$:

$$27x + 6y - z = 85$$

$$6x + 15y + 2z = 72$$

$$x + y + 54z = 110$$

Step 1: Check Diagonal Dominance

- Row 1: $|27| > |6| + |-1| = 7$ ✓
- Row 2: $|15| > |6| + |2| = 8$ ✓
- Row 3: $|54| > |1| + |1| = 2$ ✓

Step 2: Express Variables

$$x = \frac{1}{27}(85 - 6y + z)$$

$$y = \frac{1}{15}(72 - 6x - 2z)$$

$$z = \frac{1}{54}(110 - x - y)$$

Step 3: Iteration Table

Iteration (k)	x	y	z
0 (Initial)	0.0000	0.0000	0.0000
1	3.1481	3.5408	1.9132
2	2.4322	3.5720	1.9258
3	2.4257	3.5729	1.9260
4	2.4255	3.5730	1.9260

Converged Solution (4th Iterative Step):

$$x \approx 2.4255, \quad y \approx 3.5730, \quad z \approx 1.9260$$

Gauss-Seidel Example 2: Rapidly Converging System

Solve the system starting with $(x_0, y_0, z_0) = (0, 0, 0)$:

$$10x + y - z = 9$$

$$x + 10y + 2z = 27$$

$$-x + 2y + 10z = 33$$

Step 1: Express Variables

$$x^{(k+1)} = \frac{1}{10} (9 - y^{(k)} + z^{(k)})$$

$$y^{(k+1)} = \frac{1}{10} (27 - x^{(k+1)} - 2z^{(k)})$$

$$z^{(k+1)} = \frac{1}{10} (33 + x^{(k+1)} - 2y^{(k+1)})$$

Step 2: Detailed Iterations

- **Iteration 1:**

$$x_1 = \frac{1}{10} (9 - 0 + 0) = 0.9000$$

$$y_1 = \frac{1}{10} (27 - 0.9000 - 0) = 2.6100$$

$$z_1 = \frac{1}{10} (33 + 0.9000 - 2(2.6100)) = 2.8680$$

- **Iteration 2:**

$$x_2 = \frac{1}{10} (9 - 2.6100 + 2.8680) = 0.9258$$

$$y_2 = \frac{1}{10} (27 - 0.9258 - 2(2.8680)) = 2.0338$$

$$z_2 = \frac{1}{10} (33 + 0.9258 - 2(2.0338)) = 2.9858$$

- **Iteration 3 & 4:**

$$x_3 = 0.9952, \quad y_3 = 2.0033, \quad z_3 = 2.9989$$

$$x_4 = 0.9996, \quad y_4 = 2.0003, \quad z_4 = 2.9999$$

Exact Solution: $(x, y, z) = (1, 2, 3)$

Gauss-Seidel Example 3: System Requiring Rearrangement First

Solve the following system using Gauss-Seidel Method:

$$x + 2y + 10z = 19 \quad (\text{Eq. 1})$$

$$20x + y - 2z = 15 \quad (\text{Eq. 2})$$

$$3x + 20y - z = -19 \quad (\text{Eq. 3})$$

Step 1: Check Diagonal Dominance & Rearrange

Notice that initially, Eq. 1 has leading coefficient $1 < 2 + 10$. We must reorder equations to establish diagonal dominance:

- **New Row 1 (Eq. 2):** $20x + y - 2z = 15$ ($|20| > |1| + |-2|$)
- **New Row 2 (Eq. 3):** $3x + 20y - z = -19$ ($|20| > |3| + |-1|$)
- **New Row 3 (Eq. 1):** $x + 2y + 10z = 19$ ($|10| > |1| + |2|$)

Step 2: Iterative Equations

$$x = \frac{1}{20}(15 - y + 2z)$$

$$y = \frac{1}{20}(-19 - 3x + z)$$

$$z = \frac{1}{10}(19 - x - 2y)$$

Step 3: Iteration Results

Iteration	x	y	z
0 (Start)	0.0000	0.0000	0.0000
1	0.7500	-1.0625	2.0375
2	1.0069	-1.0010	1.9995
3	1.0000	-1.0000	2.0000

Exact Solution Reached in 3 Iterations:

$$(x, y, z) = (1, -1, 2)$$